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Play!! Now!

EXPLORING THE WUMPUS WORLD

A Logical Agent Adventure

PLAY

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Abstract

The Wumpus World is a classic problem in artificial intelligence, demonstrating how agents can make decisions under uncertainty. It is a grid-based environment where a rational agent must find gold while avoiding hidden dangers like pits and the Wumpus (a monster). This project explores the modeling of the environment, the agent's knowledge base, inference mechanisms, and decision-making strategies to safely and efficiently navigate the world.

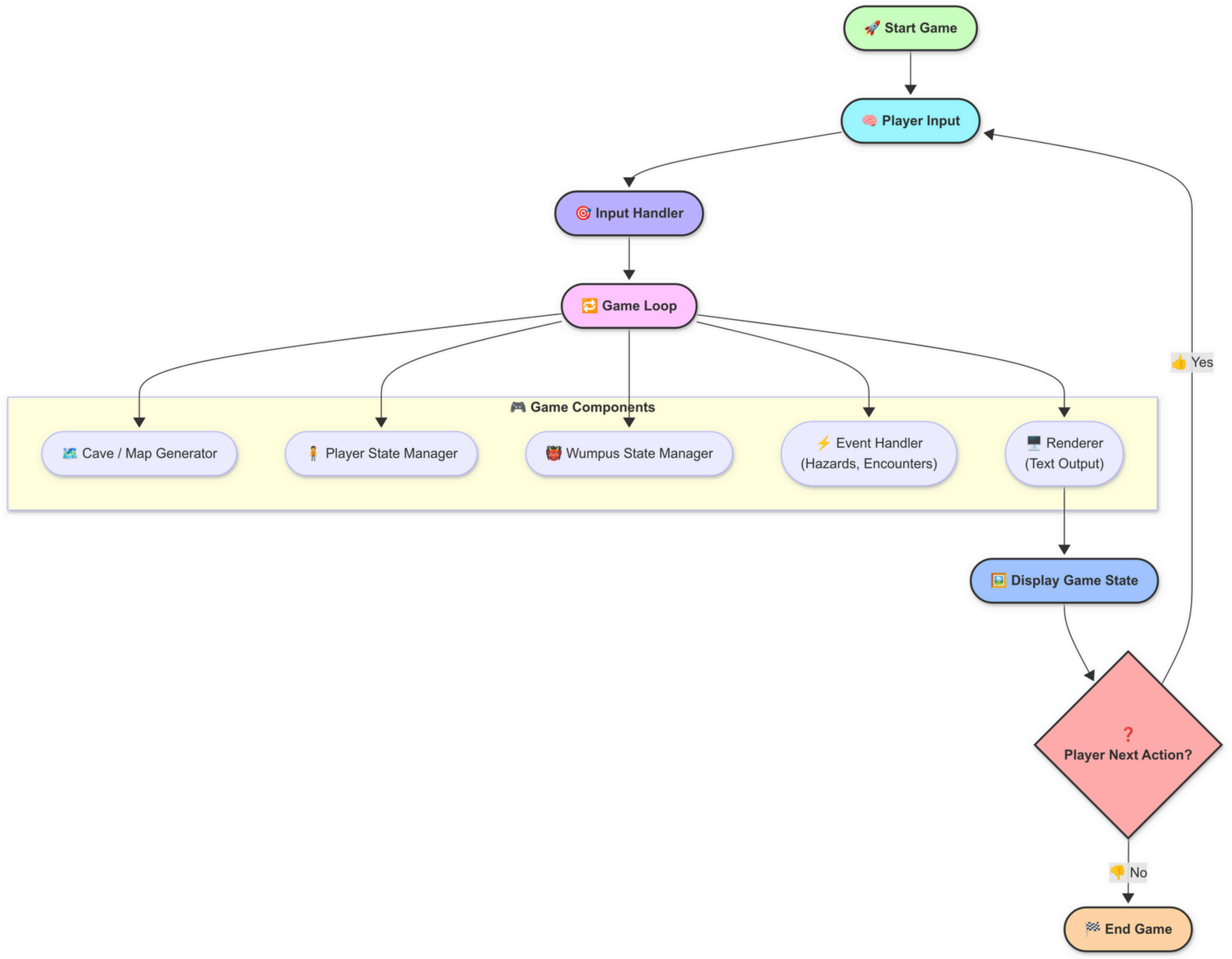
Problem Statement

In the Wumpus World, an intelligent agent must achieve two main objectives:

- Locate and retrieve the gold hidden somewhere in the environment.
- Avoid deadly hazards such as pits and the Wumpus.

The challenge lies in making decisions based on partial, uncertain information using percepts (breeze, stench, Gold, etc.). The agent must reason logically to deduce the safest moves and plan a successful strategy.

Architecture Diagram



Methodology 

Environment Design

A 4x4 or 5x5 grid with random placement of Wumpus, pits, and gold

Decision Making

- If gold percept detected (glitter) → grab.
- If stench/breeze → mark danger zones.
- If safe → move forward.

Agent Design

- Uses Percepts to update a Knowledge Base.
- Applies Logical Inference to reason about safe cells.
- Applying Backtracking

Implementation Details

Language/Tools Used:

Libraries Used in Python:

Heap

Python Enum

Queues

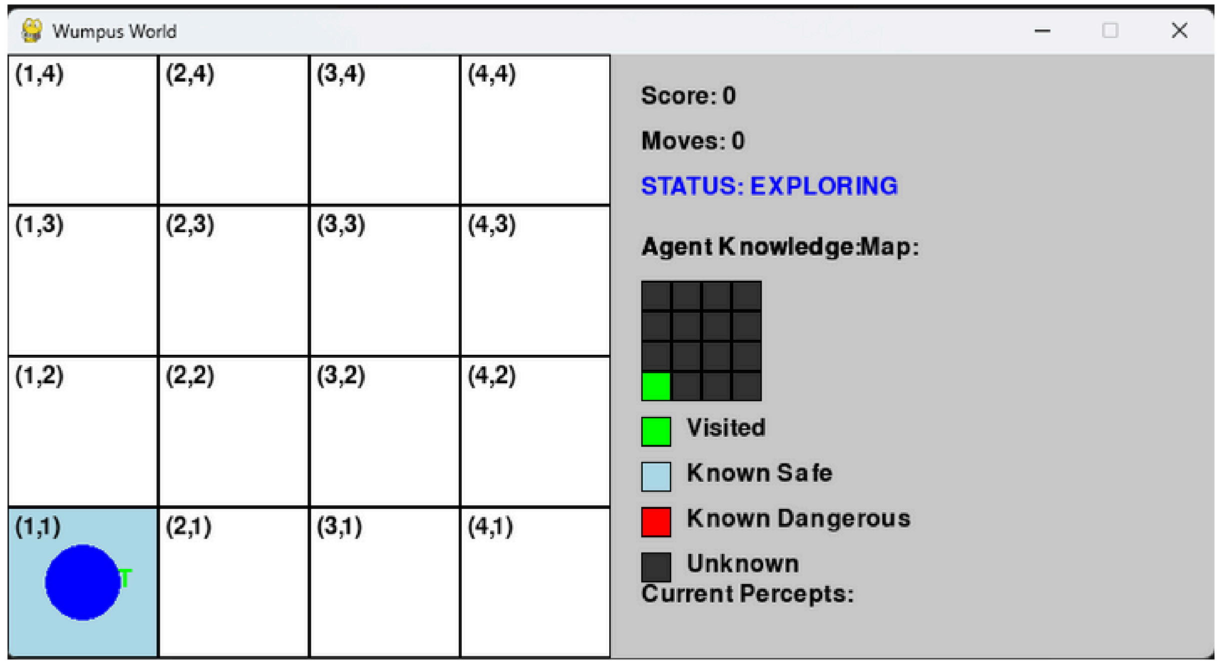
Game Rules:

- User Movement
  - Agent can move, grab
- User Skill
  - Agent can Sense pits through breeze int the adjacent block
  - Agent can Sense wumpus through strench int the adjacent block
- User Objective
  - Objective is to maximize score (gold & survival).
- User Survival Instincts
  - Agent "dies" if it falls into a pit or Wumpus eats it.

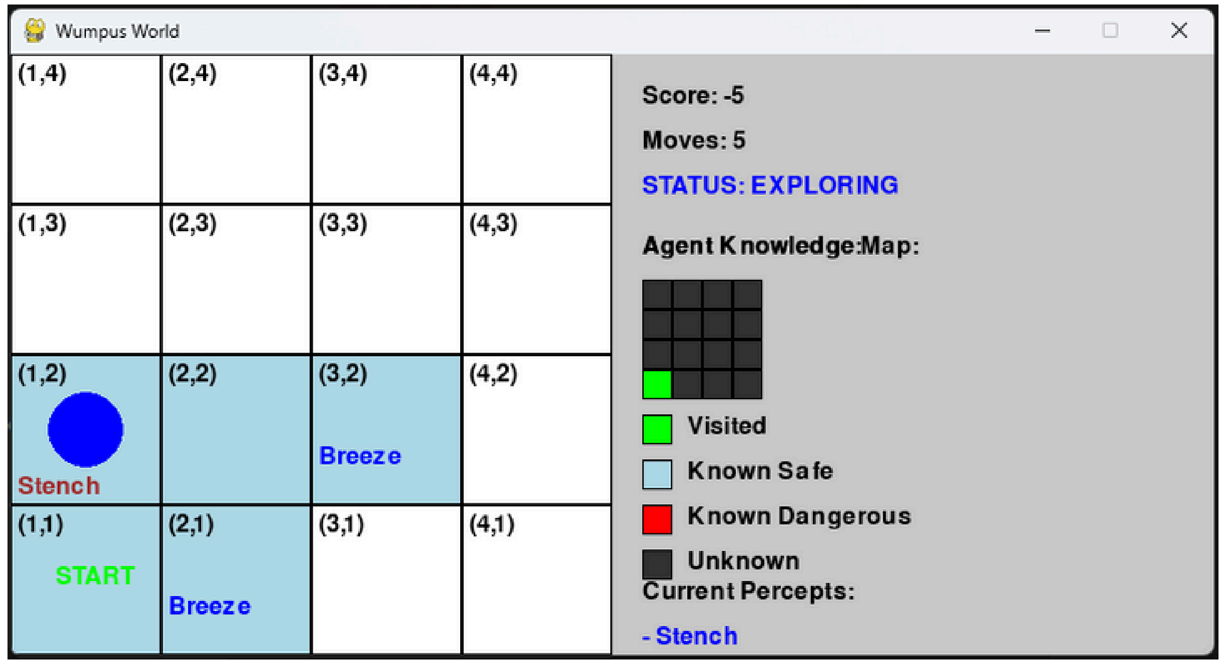
Results

Screenshots

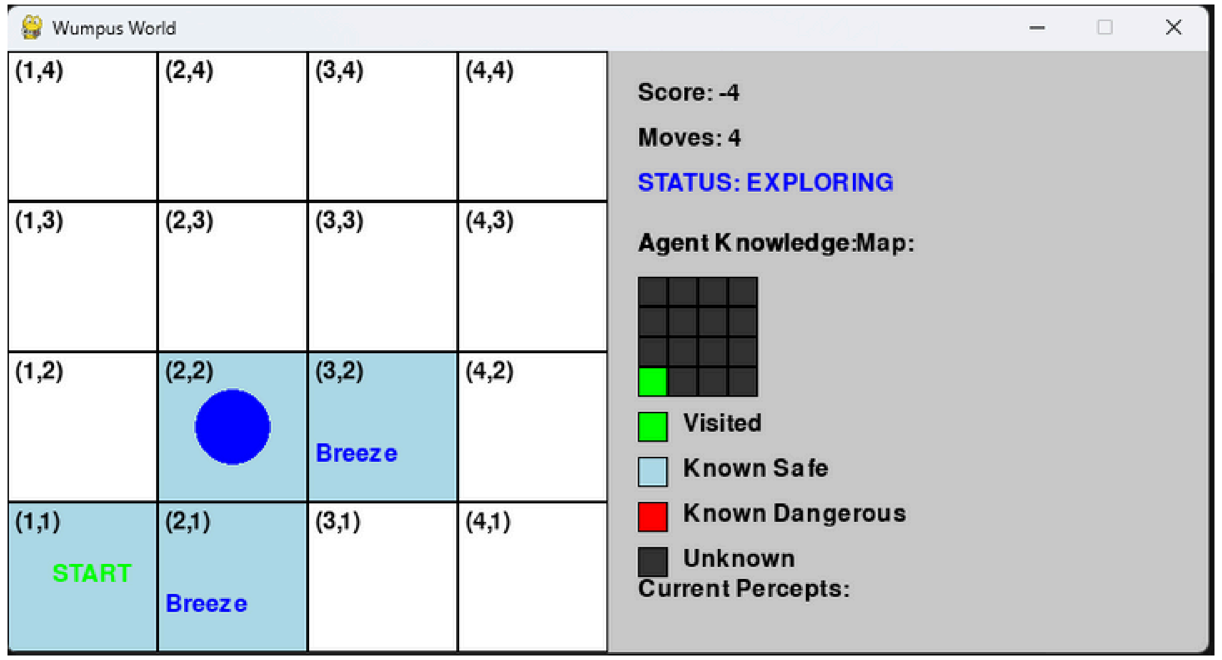
- **Start Screen**



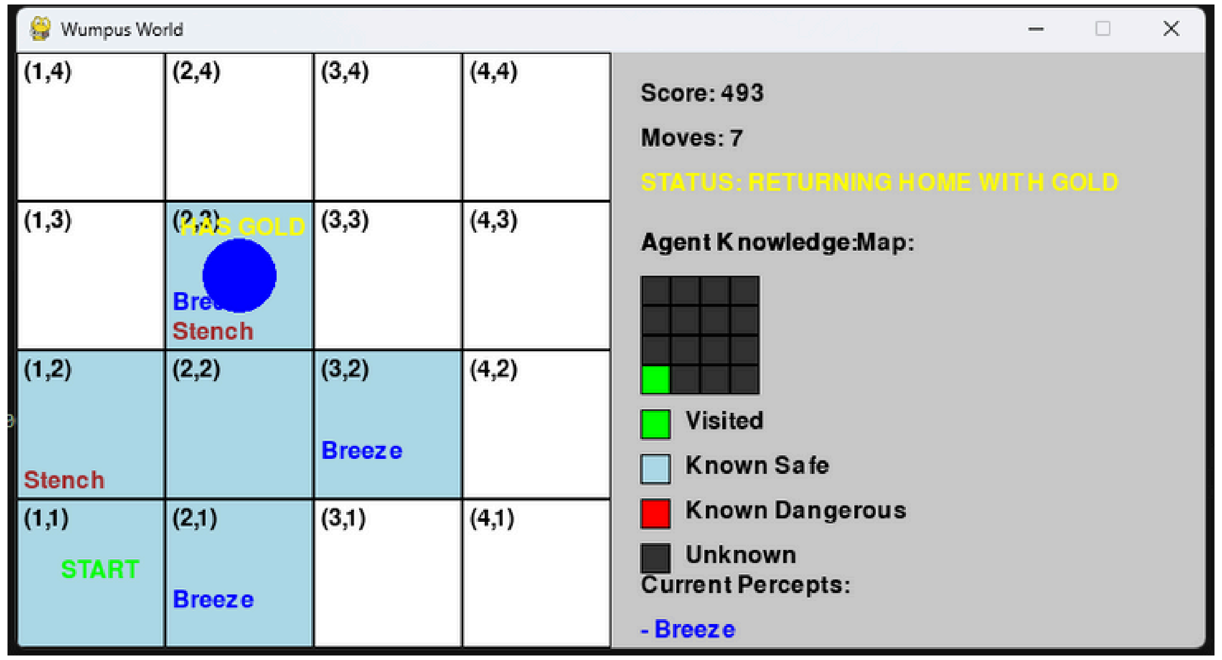
- **Strench Sensed**



- **Breeze Sensed**



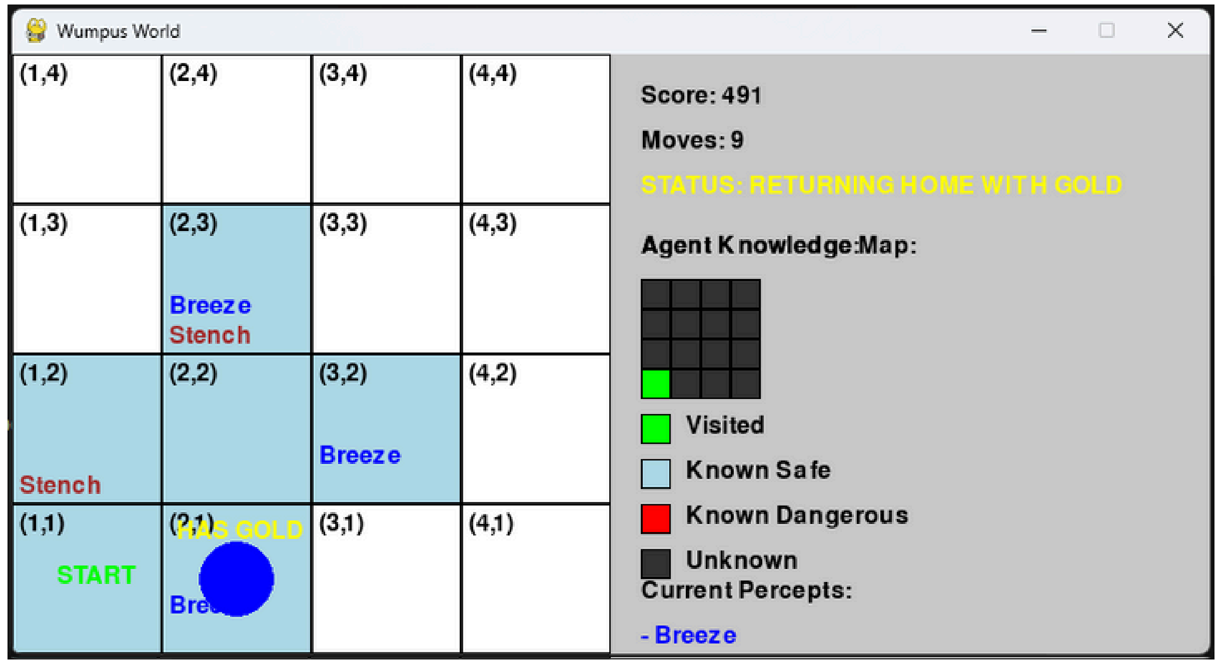
- **GOLD Grabbed**



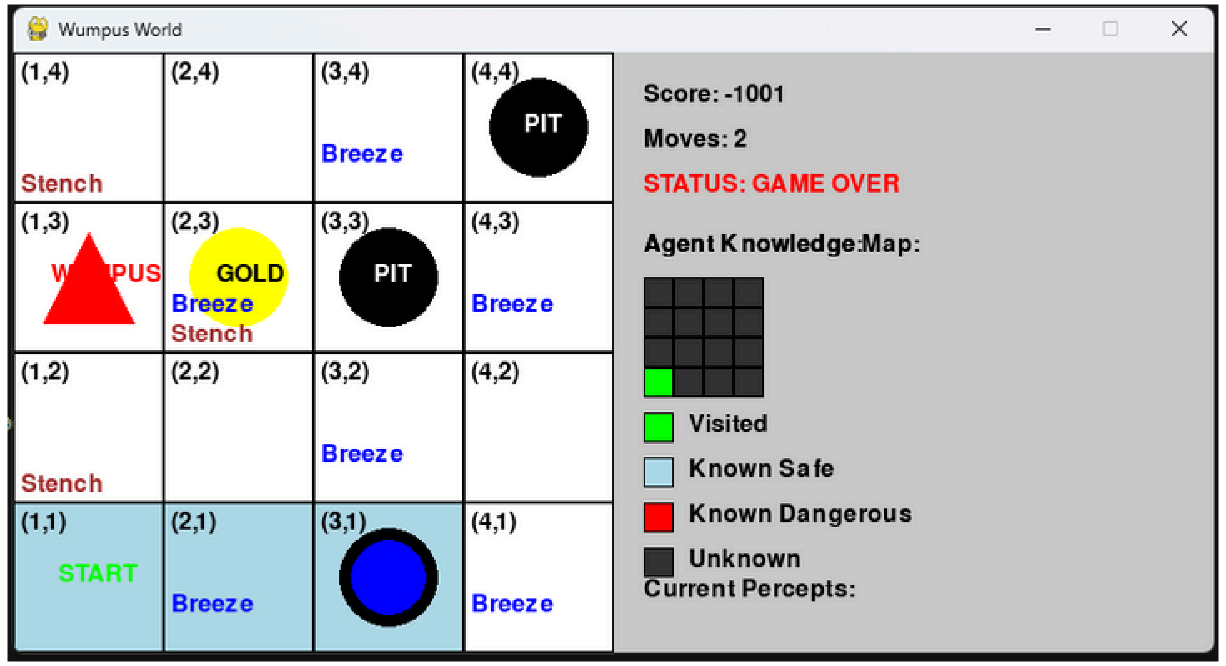
Results

Screenshots

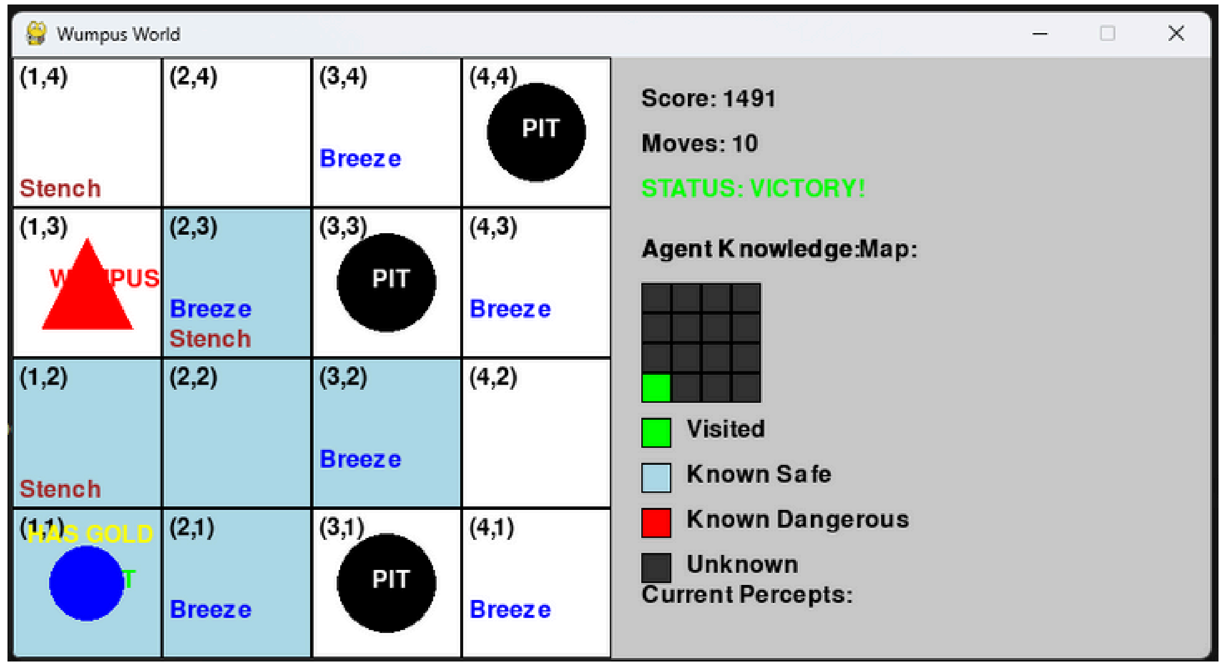
- **Back-Tracking**



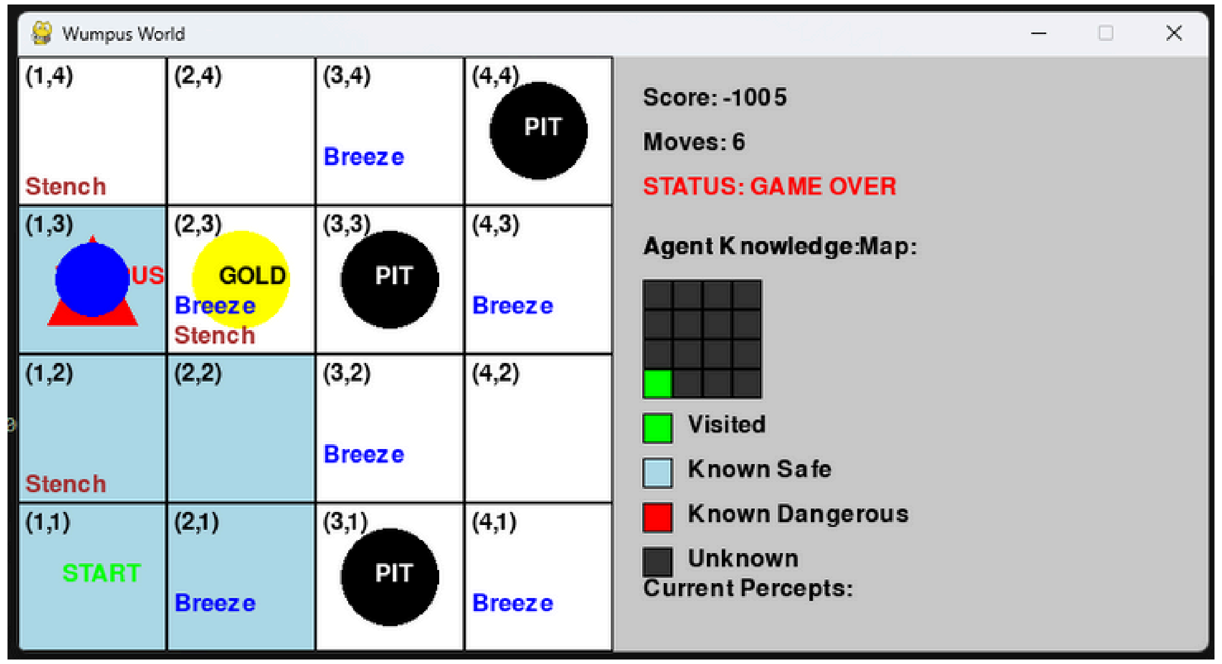
- **Fall in PIT(Game Over)**



- **Game Won**



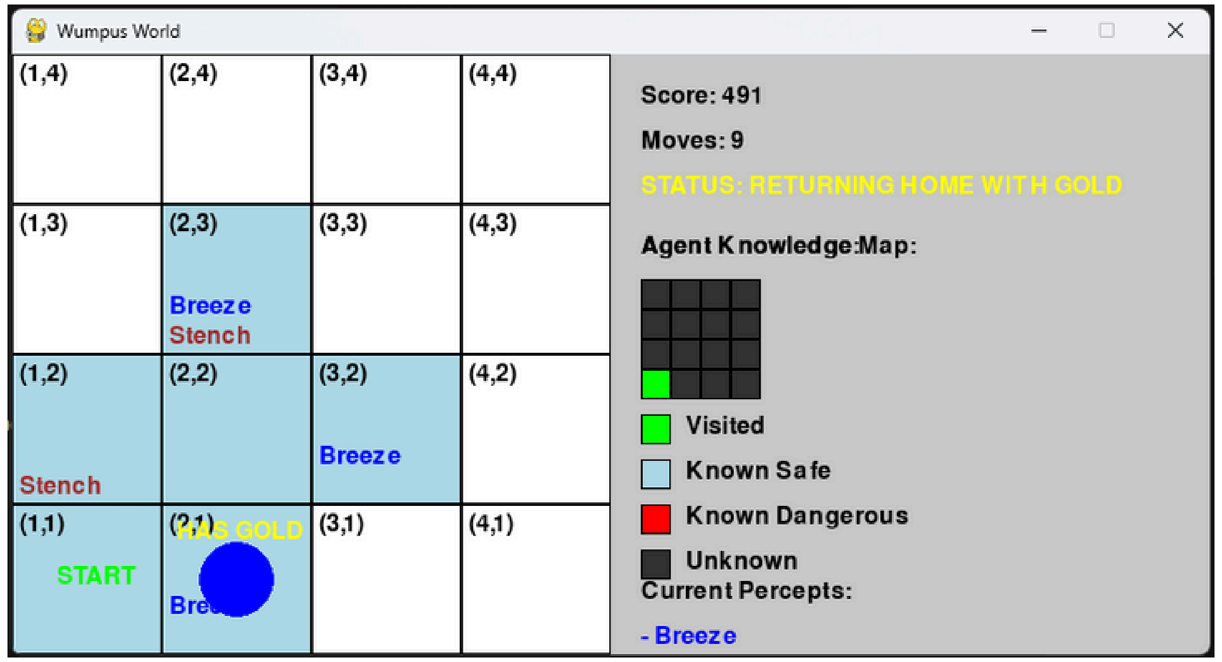
- **Eaten by Wumpus(Game Over)**



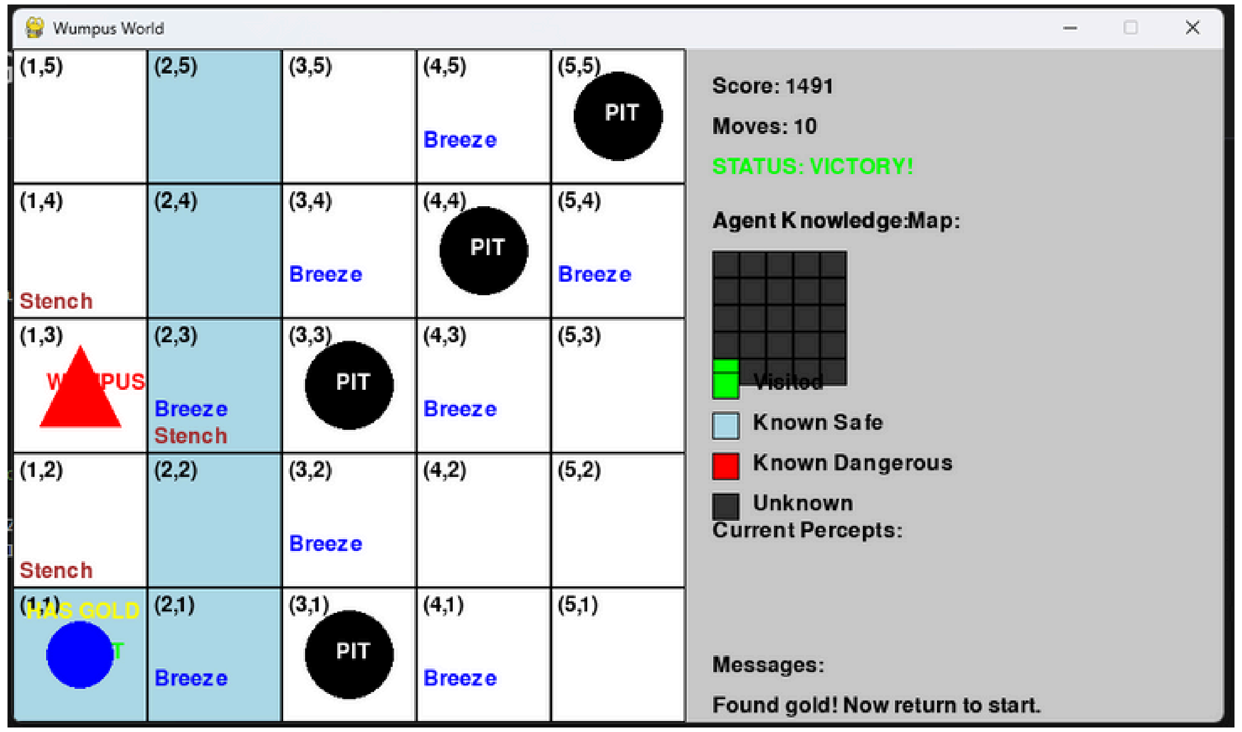
Results

Screenshots

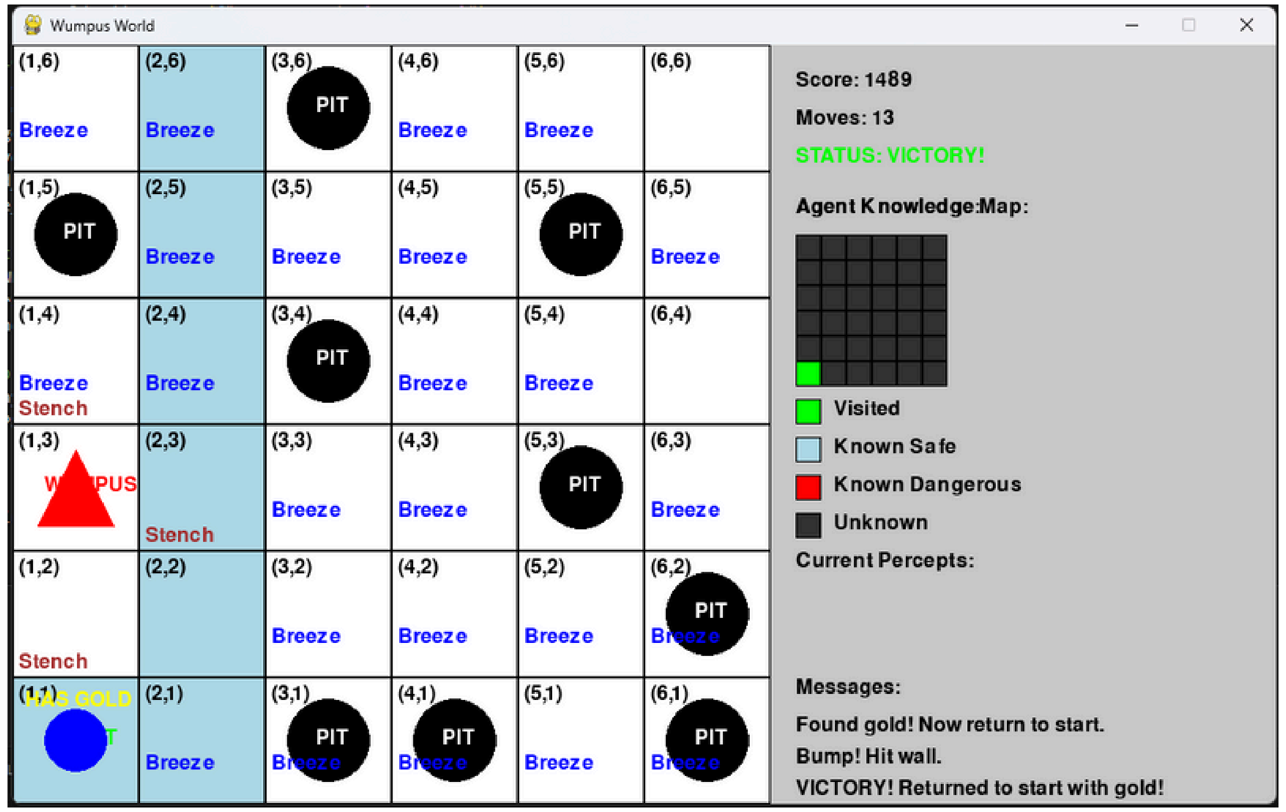
Different Levels



LEVEL 1 : 4 X4



LEVEL 2 : 5 X 5



LEVEL 3 : 6 X 6

Many More!

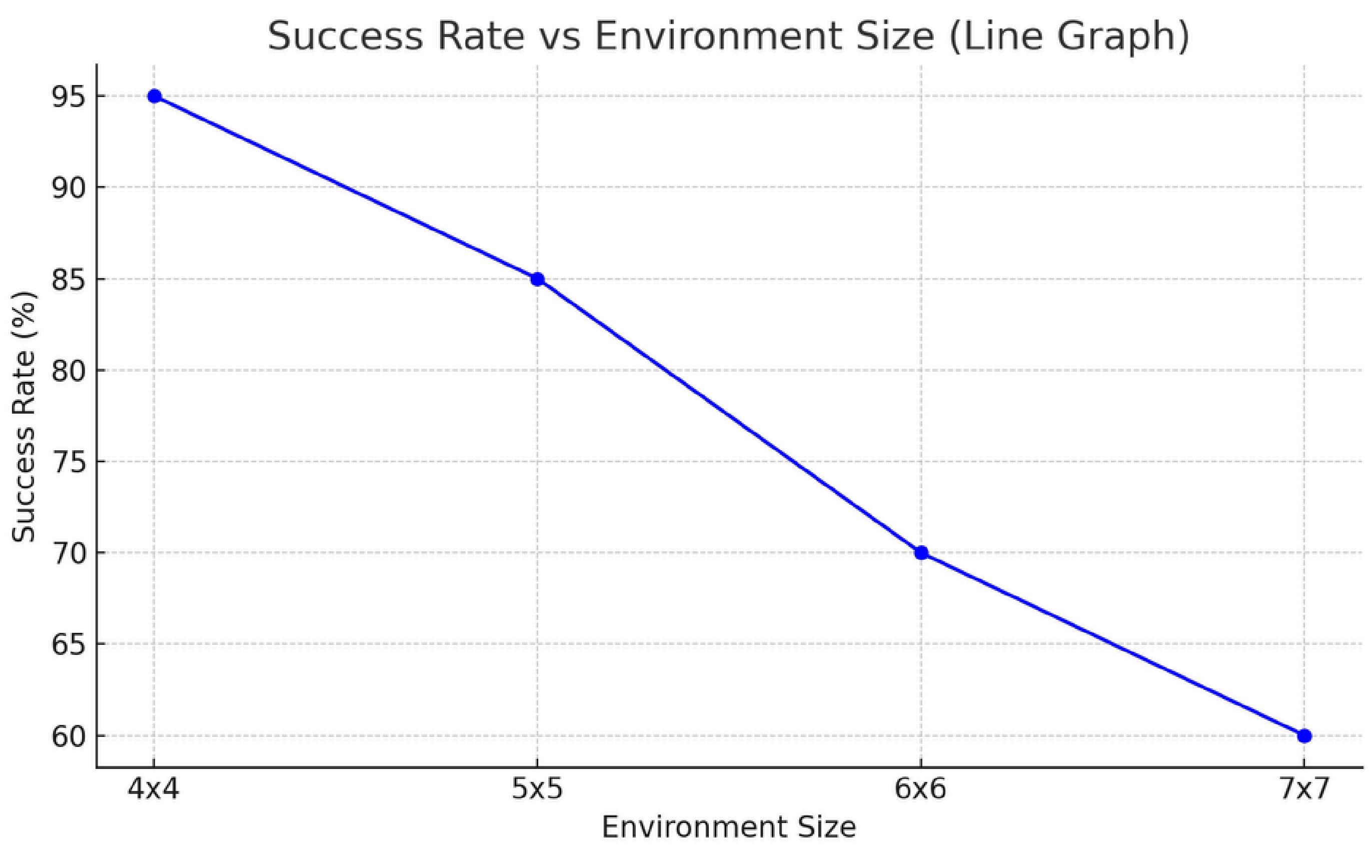
Coming Soon!!

Results

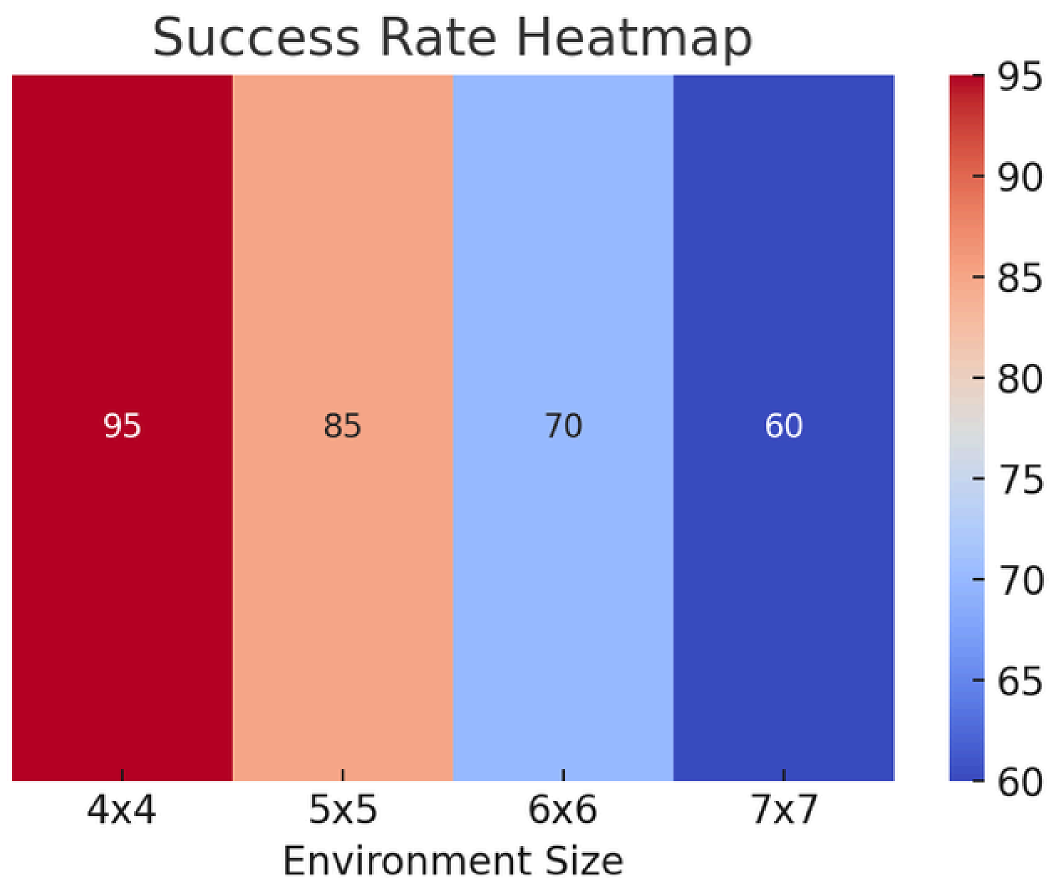
Comparison graphs

Environment Size	Success Rate (%)
4x4	95%
5x5	85%
6x6	70%
7x7	60%

Line Graph



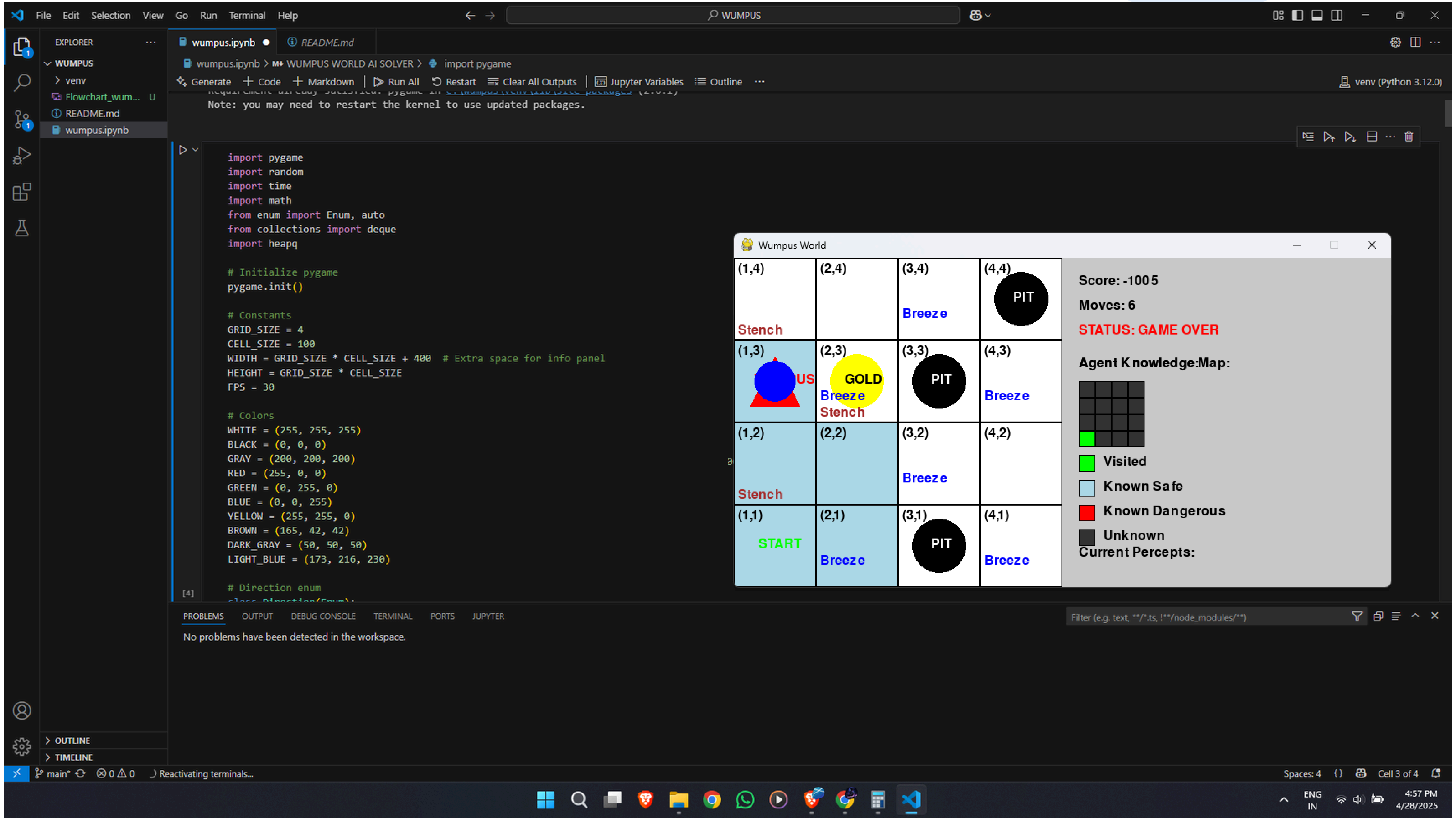
Heat Map/ Confusion Graph

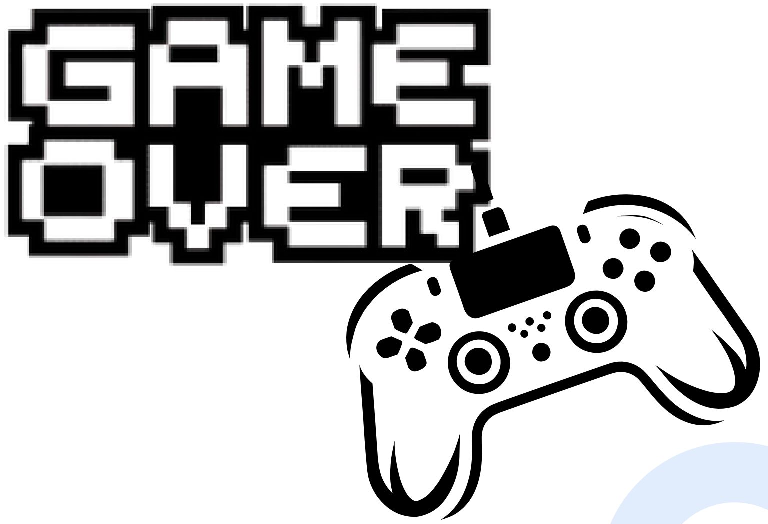


Results

DEMO VIDEO

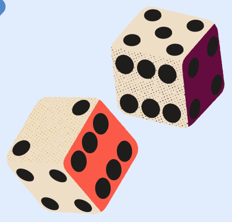
Click To Check the Demo Video





Conclusion

The project successfully demonstrated an intelligent agent navigating the Wumpus World, avoiding hazards while retrieving gold. Performance was affected by grid size and hazard complexity, with larger grids leading to lower success rates. Future improvements include using probabilistic reasoning and machine learning to enhance decision-making and agent adaptability in dynamic environments.



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THANK YOU!

MADE BY
AC1
MADE BY

Piyush Saini (RA2311026010105)

Vansh Agrawal (RA2311026010120)

Soubhik Sadhu (RA2311026010122)